University of Hertfordshire UH

School of Physics, Engineering and Computer Science

MSc Data Science Project
7PAM2002
Department of Physics, Astronomy and Mathematics

Data Science FINAL PROJECT REPORT

Project Title:

Classification of mushroom edibility based on morphological characteristics using machine learning

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Date Submitted: 22-04-2026
Word Count: 5493
GITHUB LINK:
https://github.com/satwik1704/mushroomclassification

DECLARATION STATEMENT

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ABBREVIATIONS

Abbreviation	Full Form
ML	Machine Learning
AI	Artificial Intelligence
SVM	Support Vector Machine
RF	Random Forest
DT	Decision Tree
XGBoost	Extreme Gradient Boosting
KNN	K-Nearest Neighbors
CSV	Comma Separated Values
EDA	Exploratory Data Analysis
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative
F1-score	Harmonic Mean of Precision and Recall
API	Application Programming Interface

CHAPTER I

INTRODUCTION

1.1 Background

Mushrooms are a diverse group of fungi that play an essential role in ecosystems and human nutrition. They are widely consumed due to their rich content of proteins, vitamins, minerals, and antioxidants. However, not all mushrooms are safe to eat. Some species contain toxic compounds that can cause severe illness or even death. The challenge of distinguishing edible mushrooms from poisonous ones has been a long-standing problem in both biology and public health.

Traditionally, mushroom identification relies on expert knowledge and manual inspection of morphological features such as cap shape, color, gill size, odor, and habitat. While experienced mycologists can accurately identify species, this expertise is not easily accessible to the general public. Moreover, many poisonous mushrooms closely resemble edible varieties, making manual identification unreliable and risky.

With the advancement of technology, machine learning has emerged as a powerful tool for classification problems. Machine learning algorithms can analyze large datasets and identify patterns that may not be obvious to humans. In the context of mushroom classification, machine learning can be used to analyze morphological characteristics and predict whether a mushroom is edible or poisonous.

This project focuses on developing a machine learning-based system for classifying mushroom edibility. The system uses a dataset containing various morphological attributes of mushrooms and applies multiple machine learning algorithms to achieve accurate classification. By leveraging computational techniques, the study aims to provide a reliable and efficient solution to a critical real-world problem.

1.2 Problem Statement

The primary problem addressed in this study is the difficulty in accurately identifying edible and poisonous mushrooms based on their morphological characteristics. Manual identification methods are prone to errors due to the following reasons:

- Many poisonous mushrooms closely resemble edible ones

- Identification requires expert knowledge that is not widely available
- Environmental factors can alter mushroom appearance
- Misidentification can lead to serious health consequences

Existing traditional methods are insufficient for ensuring safety, especially for individuals without formal training in mycology. Therefore, there is a need for an automated system that can accurately classify mushrooms based on observable features.

The problem can be formally defined as a classification task where the goal is to predict whether a mushroom is edible or poisonous using its morphological attributes. The challenge lies in handling categorical data, selecting relevant features, and choosing appropriate machine learning models to achieve high accuracy.

1.3 Justification of the Study

This study is justified by several important factors related to public health, technological advancement, and practical applications.

First, food safety is a critical concern worldwide. Mushroom poisoning incidents are reported frequently, often due to misidentification. An automated classification system can significantly reduce such risks by providing accurate predictions.

Second, the use of machine learning in biological classification is an emerging field. Applying machine learning techniques to mushroom classification demonstrates the potential of artificial intelligence in solving real-world problems in biology and agriculture.

Third, the availability of structured datasets and powerful computational tools makes it feasible to develop accurate classification models. This study leverages these resources to build a reliable system.

Fourth, the proposed system can be integrated into mobile or web applications, making it accessible to a wide range of users. Such applications can assist farmers, researchers, and the general public in identifying mushrooms safely.

Finally, this study contributes to academic research by comparing different machine learning algorithms and analyzing their performance in a real-world classification problem.

1.4 Research Questions

The study is guided by the following research questions:

1. How can morphological characteristics of mushrooms be used to classify their edibility?
2. Which machine learning algorithms are most effective for mushroom classification?
3. How does data preprocessing affect the performance of classification models?
4. What is the impact of feature selection on model accuracy?
5. How can the developed model be applied in real-world scenarios?

These questions aim to explore both the theoretical and practical aspects of mushroom classification using machine learning.

1.5 Aims and Objectives

The main aim of this project is to develop a machine learning-based system for classifying mushrooms as edible or poisonous using morphological characteristics.

The specific objectives of the study are as follows:

- To collect and preprocess a dataset containing mushroom attributes
- To analyze the dataset and identify important features
- To implement various machine learning algorithms for classification
- To evaluate model performance using appropriate metrics
- To optimize models using hyperparameter tuning
- To compare different algorithms and identify the best-performing model
- To propose a system that can be used for real-world applications

DATA FLOW DIAGRAM

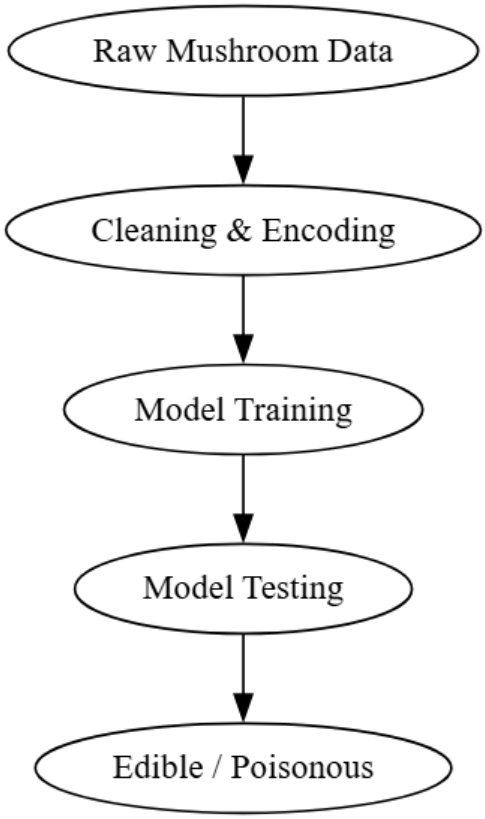

Figure 1.1: Data Flow Diagram for Mushroom Classification System

This diagram illustrates the data flow in the mushroom classification system. It shows how raw mushroom data is processed through cleaning and encoding, followed by model training and testing, ultimately leading to classification of mushrooms as edible or poisonous.

TABLE 1.1: Key Morphological Features

Feature Name	Description
Cap Shape	Shape of mushroom cap
Cap Color	Color of the cap
Odor	Smell of the mushroom
Gill Size	Size of gills
Gill Color	Color of gills
Stalk Shape	Shape of the stalk

Feature Name	Description
Habitat	Natural growing environment

TABLE 1.2: Machine Learning Algorithms Used

Algorithm	Purpose
Random Forest	Ensemble learning method
Support Vector Machine	High-dimensional classification
XGBoost	Boosting-based model

CHAPTER 2

REVIEW OF LITERATURE

2.1 Introduction

The classification of mushrooms into edible and poisonous categories has been a subject of significant interest in both biological sciences and computational research. With the increasing availability of structured datasets and advancements in machine learning techniques, researchers have explored automated methods for identifying mushroom edibility based on morphological characteristics.

This chapter presents a comprehensive review of existing literature related to mushroom classification using machine learning. It covers traditional identification methods, the evolution of machine learning techniques, data preprocessing strategies, feature selection approaches, and the application of various classification algorithms. Additionally, it discusses recent research works and highlights gaps that this study aims to address.

2.2 Traditional Methods of Mushroom Classification

Before the advent of machine learning, mushroom classification relied heavily on manual inspection and expert knowledge. Mycologists used observable features such as cap shape, color, texture, gill structure, odor, and habitat to identify mushroom species.

Traditional classification methods include:

- Visual inspection of morphological features
- Use of field guides and taxonomic keys
- Microscopic examination of spores
- Chemical testing for toxins

While these methods are effective when performed by experts, they have several limitations. They require extensive training and experience, are time-consuming, and are prone to human error. Moreover, many poisonous mushrooms closely resemble edible ones, making accurate identification difficult.

2.3 Emergence of Machine Learning in Biological Classification

Machine learning has revolutionized the field of biological classification by enabling automated analysis of complex datasets. It allows systems to learn patterns from data and make predictions without explicit programming.

In mushroom classification, machine learning models analyze morphological attributes and classify mushrooms based on learned patterns. The advantages of using machine learning include:

- High accuracy in classification tasks
- Ability to handle large datasets
- Reduced reliance on expert knowledge
- Faster and scalable solutions

Researchers have applied various supervised learning algorithms to classify mushrooms, demonstrating the effectiveness of computational approaches in biological problems.

2.4 Dataset Utilization in Mushroom Classification

One of the most widely used datasets in mushroom classification research is the UCI Mushroom Dataset. This dataset contains thousands of instances with categorical attributes describing mushroom characteristics.

Key features of the dataset include:

- Completely categorical data
- No missing values in most attributes
- Balanced distribution of edible and poisonous classes
- Multiple descriptive features such as odor, cap color, and habitat

Proper dataset handling is crucial for building accurate models. Researchers emphasize the importance of data preprocessing, including encoding categorical variables and handling inconsistencies.

TABLE 2.1: Characteristics of Mushroom Dataset

Attribute Type	Description
Number of Features	20+ categorical attributes
Target Variable	Edible or Poisonous
Data Type	Categorical
Dataset Size	Large (8000+ instances)
Missing Values	Minimal or encoded as categories

2.5 Data Preprocessing Techniques

Data preprocessing is a critical step in machine learning that significantly impacts model performance. Since mushroom datasets are primarily categorical, preprocessing involves transforming data into a suitable numerical format.

Common preprocessing techniques include:

- Label Encoding
- One-Hot Encoding
- Handling missing values
- Feature scaling (if required)
- Removing redundant features

Encoding techniques convert categorical values into numerical representations that machine learning algorithms can process. One-hot encoding is particularly useful as it avoids introducing ordinal relationships between categories.

DATA PREPROCESSING PIPELINE

2.6 Feature Selection and Importance

Feature selection plays a vital role in improving model performance by identifying the most relevant attributes for classification. Not all features contribute equally to the prediction of mushroom edibility.

Studies have shown that certain features such as odor, gill color, and spore print color are highly significant in distinguishing edible mushrooms from poisonous ones.

Feature selection methods include:

- Filter methods (correlation-based)
- Wrapper methods (recursive feature elimination)
- Embedded methods (feature importance in tree-based models)

By reducing irrelevant features, models become more efficient and less prone to overfitting.

TABLE 2.2: Important Features in Mushroom Classification

Feature Name	Importance Level
Odor	High
Gill Color	High
Spore Print Color	Medium
Cap Shape	Medium
Habitat	Low

2.7 Machine Learning Algorithms for Classification

Various machine learning algorithms have been applied to mushroom classification. Each algorithm has its strengths and limitations.

2.7.1 Decision Tree

Decision trees are simple and interpretable models that split data based on feature values. They are widely used due to their ease of implementation and visualization.

Advantages:

- Easy to understand

- Handles categorical data well

Limitations:

- Prone to overfitting

2.7.2 Random Forest

Random forest is an ensemble method that combines multiple decision trees to improve accuracy and reduce overfitting.

Advantages:

- High accuracy
- Robust to noise

Limitations:

- Increased computational complexity

2.7.3 Support Vector Machine

Support Vector Machines (SVM) are effective in high-dimensional spaces and can handle complex decision boundaries.

Advantages:

- Effective in complex datasets
- Works well with clear margins

Limitations:

- Requires careful parameter tuning

2.7.4 Gradient Boosting and XGBoost

Boosting algorithms such as XGBoost improve performance by combining weak learners sequentially.

Advantages:

- High predictive power

- Handles complex relationships

Limitations:

- Computationally intensive

MODEL COMPARISON FLOW

2.8 Evaluation Metrics in Classification

To assess the performance of machine learning models, several evaluation metrics are used:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

These metrics provide a comprehensive understanding of model performance, especially in classification tasks.

TABLE 2.3: Evaluation Metrics

Metric	Description
Accuracy	Overall correctness of the model
Precision	Correct positive predictions
Recall	Ability to identify actual positives
F1-score	Balance between precision and recall

2.9 Challenges in Mushroom Classification

Despite advancements, several challenges remain in mushroom classification:

- High similarity between edible and poisonous species
- Handling categorical data effectively
- Risk of overfitting in complex models
- Need for real-time prediction systems

These challenges motivate further research and improvement in classification techniques.

2.10 Recent Advances in Machine Learning Applications

Recent studies have explored advanced techniques such as deep learning and hybrid models. However, traditional machine learning models often outperform deep learning in structured datasets like mushroom classification.

Researchers are also focusing on:

- Model interpretability
- Feature importance analysis
- Deployment in mobile applications

These advancements aim to make classification systems more accessible and reliable.

2.11 Related Works

Several researchers have contributed to the field of mushroom classification using machine learning.

- Studies using Decision Trees achieved high accuracy but faced overfitting issues
- Random Forest models demonstrated improved performance and robustness
- SVM-based approaches showed effectiveness in handling complex feature spaces
- XGBoost models achieved state-of-the-art accuracy due to boosting techniques

Some studies reported near 100% accuracy, indicating that mushroom classification is highly suitable for machine learning applications when proper preprocessing is applied.

TABLE 2.4: Summary of Related Works

Study Approach	Algorithm Used	Accuracy Level
Traditional ML	Decision Tree	High
Ensemble Learning	Random Forest	Very High
Kernel Methods	SVM	High
Boosting	XGBoost	Very High

CHAPTER 3

RESEARCH METHOD

3.1 OVERVIEW

This chapter presents the methodology used to develop the mushroom classification system. The research adopts a systematic machine learning approach that involves data collection, preprocessing, feature engineering, model selection, training, evaluation, and optimization.

The primary objective of this methodology is to build an accurate and reliable model that can classify mushrooms as edible or poisonous based on their morphological characteristics. The process is designed to ensure reproducibility, efficiency, and scalability.

The workflow consists of multiple stages, starting from raw data input and ending with model prediction. Each stage is carefully implemented to enhance performance and minimize errors.

3.2 RESEARCH DESIGN

The research design follows a supervised learning framework where labeled data is used to train classification models. The dataset contains input features representing mushroom characteristics and a target variable indicating whether the mushroom is edible or poisonous.

Key components of the research design include:

- Structured dataset with categorical features
- Supervised classification approach
- Multiple algorithm comparison
- Performance evaluation using standard metrics

The study adopts an experimental approach where different machine learning models are trained and evaluated under the same conditions to determine the best-performing model.

TABLE 3.1: Research Design Summary

Component	Description
Approach	Supervised Learning
Data Type	Categorical
Task	Classification
Target Variable	Edible / Poisonous
Algorithms Used	Random Forest, SVM, XGBoost

3.3 DATA COLLECTION AND DESCRIPTION

The dataset used in this study is a structured mushroom dataset containing various morphological attributes. Each record represents a mushroom instance with features describing its physical characteristics.

Dataset characteristics:

- Number of instances: Large dataset (8000+ samples)
- Number of features: 20+ categorical attributes
- Target classes: Edible (e) and Poisonous (p)

Example features:

- Cap shape
- Cap color
- Odor
- Gill size
- Gill color
- Stalk shape
- Habitat

The dataset is suitable for classification tasks due to its structured format and clear labeling.

3.3.1 DATA PREPROCESSING

Data preprocessing is a crucial step in the machine learning pipeline. Since the dataset consists entirely of categorical features, it must be converted into a numerical format before training models.

Steps involved in preprocessing:

1. Data Cleaning
    - Removal of inconsistencies
    - Verification of dataset integrity
2. Handling Missing Values
    - Missing values treated as separate categories or removed
3. Encoding Categorical Variables
    - Label Encoding for ordinal features
    - One-Hot Encoding for nominal features
4. Feature Selection
    - Removal of irrelevant or redundant features
5. Train-Test Split
    - Dataset divided into training and testing sets

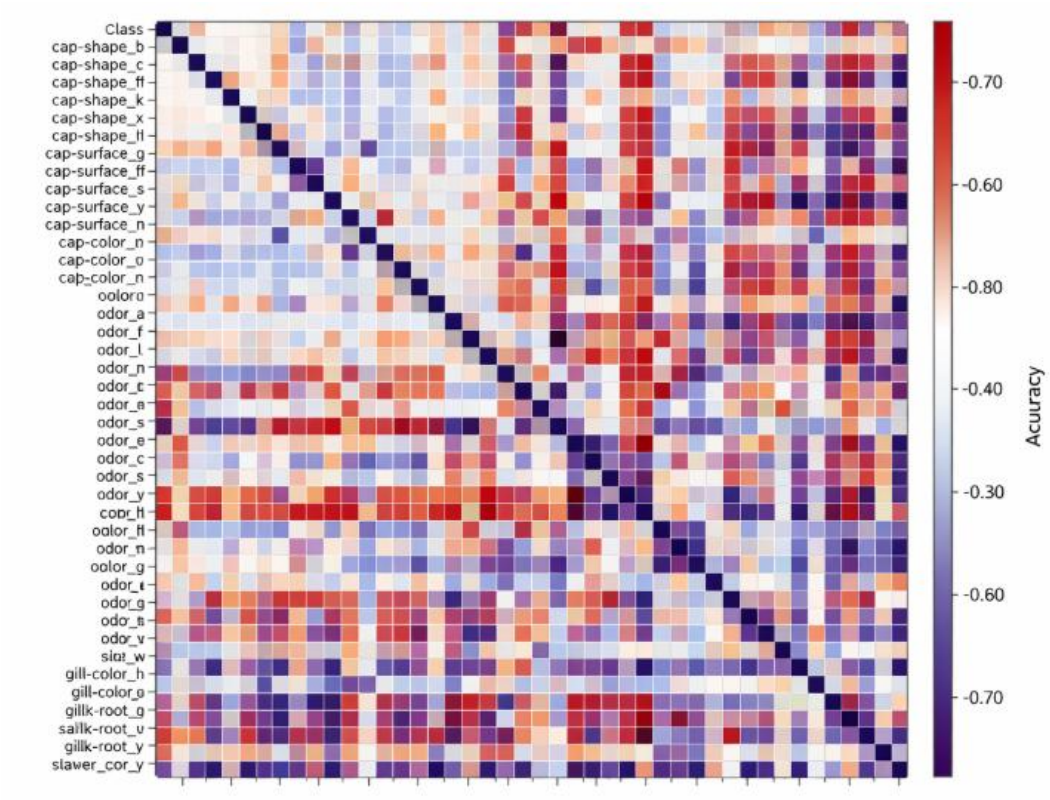

Figure 3.3: Feature Correlation Heatmap of Encoded Mushroom Dataset

The correlation heatmap illustrates the relationship between encoded mushroom features and the target variable. It highlights that features such as odor, gill color, and spore print color have strong influence on classification. This validates the effectiveness of feature encoding and supports model learning.

DATA PREPROCESSING PIPELINE

3.4 MODEL DEVELOPMENT

In this study, multiple machine learning algorithms are implemented to classify mushrooms. The models are trained using the processed dataset and evaluated based on their performance.

Models used:

- Decision Tree
- Random Forest

- Support Vector Machine
- XGBoost

Each model is trained separately and compared using evaluation metrics.

TABLE 3.2: Model Configuration

Model	Key Characteristics
Random Forest	Ensemble of multiple trees
SVM	High-dimensional classifier
XGBoost	Boosting-based ensemble model

3.4.1 MODEL TRAINING PROCESS

The model training process involves feeding the training dataset into machine learning algorithms to learn patterns and relationships.

Steps in training:

- Input encoded dataset
- Initialize model with parameters
- Train model using training data
- Validate model using test data

The training process ensures that the model learns to distinguish between edible and poisonous mushrooms effectively.

MODEL TRAINING FLOW

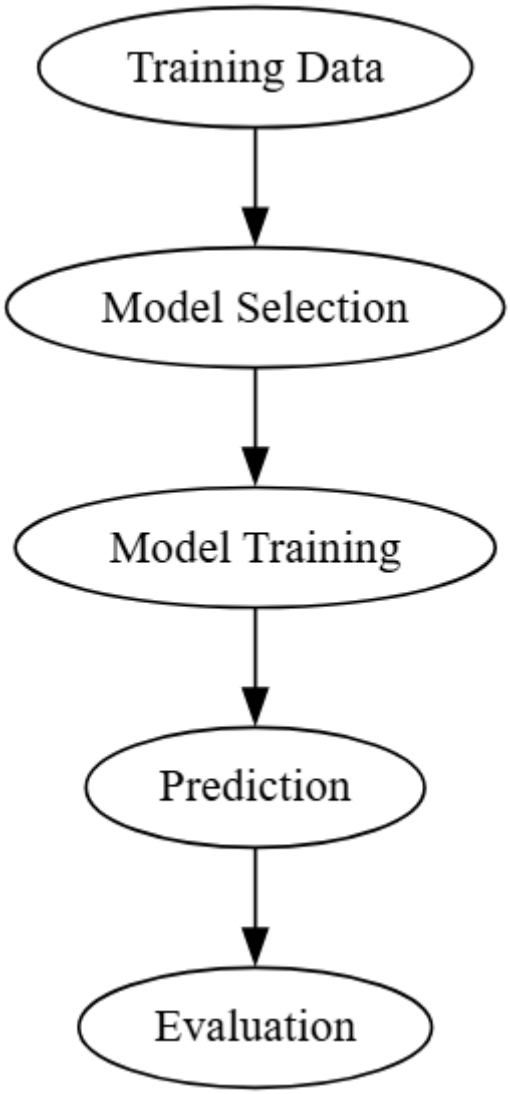

Figure 3.6: Machine Learning Model Training and Evaluation Workflow

This diagram illustrates the workflow of model development, starting from training data and model selection, followed by model training, prediction, and evaluation. It represents the core process used to build and assess machine learning models for accurate mushroom classification.

3.5 MODEL EVALUATION

Model evaluation is performed to assess the performance of each classification algorithm. The evaluation is based on standard metrics used in classification problems.

Evaluation metrics used:

- Accuracy

- Precision
- Recall
- F1-score
- Confusion Matrix

These metrics help in understanding the strengths and weaknesses of each model.

TABLE 3.3: Evaluation Metrics Description

Metric	Purpose
Accuracy	Overall correctness
Precision	Correct positive predictions
Recall	Detection of actual positives
F1-score	Balance of precision and recall

3.5.1 CONFUSION MATRIX ANALYSIS

The confusion matrix provides a detailed breakdown of classification results.

It includes:

- True Positives
- True Negatives
- False Positives
- False Negatives

This helps in identifying misclassification patterns and improving model performance.

CONFUSION MATRIX FLOW

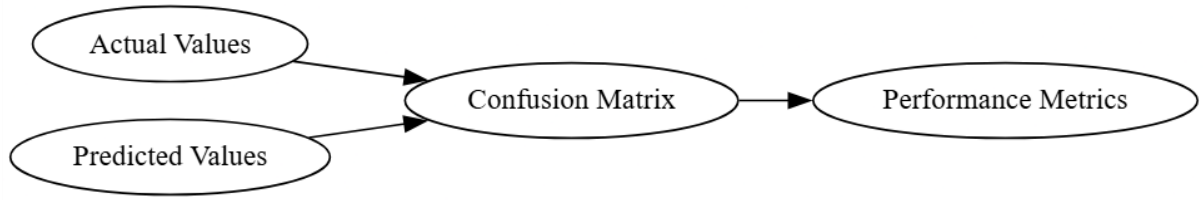

Figure 3.5: Confusion Matrix Evaluation Flow

This diagram illustrates the process of evaluating model performance using a confusion matrix. It shows how actual values and predicted values are compared to generate the confusion matrix, which is then used to compute performance metrics such as accuracy, precision, recall, and F1-score.

3.6 HYPERPARAMETER TUNING

Hyperparameter tuning is used to optimize model performance by selecting the best combination of parameters.

Techniques used:

- Grid Search
- Cross-Validation

Parameters tuned:

- Number of trees (Random Forest)
- Kernel type (SVM)
- Learning rate (XGBoost)

Tuning helps improve accuracy and reduce overfitting.

HYPERPARAMETER TUNING PROCESS

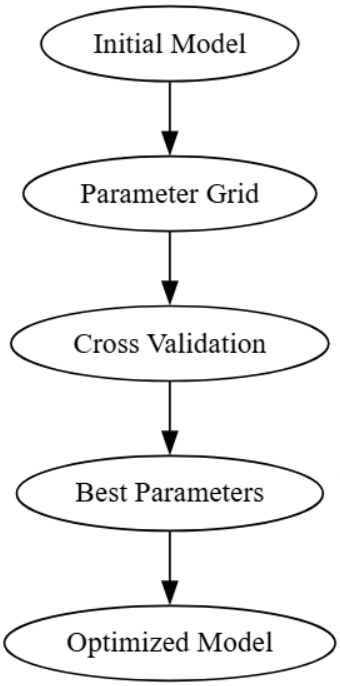

Figure 3.6: Hyperparameter Tuning Process Using Grid Search and Cross-Validation

This diagram illustrates the hyperparameter tuning workflow, where an initial model is optimized using a parameter grid and cross-validation. The process evaluates multiple parameter combinations to identify the best configuration, resulting in an optimized model with improved performance and reduced overfitting.

3.7 IMPLEMENTATION TOOLS AND TECHNOLOGIES

The implementation of the project is carried out using modern tools and programming environments.

Tools used:

- Programming Language: Python
- Libraries:
  - NumPy
  - Pandas
  - Scikit-learn
  - XGBoost
- Development Environment: Jupyter Notebook

The models were trained and saved as serialized files for future use.

You can access the implementation notebook here:

TABLE 3.4: Tools and Libraries

Tool/Library	Purpose
Python	Programming language
Pandas	Data manipulation
NumPy	Numerical operations
Scikit-learn	Machine learning models
XGBoost	Boosting algorithm

3.8 SYSTEM IMPLEMENTATION WORKFLOW

The overall workflow integrates all stages of the research into a single pipeline.

SYSTEM WORKFLOW DIAGRAM

Figure 3.8: System Implementation Workflow for Mushroom Classification

This diagram represents the overall workflow of the mushroom classification system. It shows the sequential stages from data input and preprocessing to feature engineering, model training, evaluation, and final prediction, illustrating the complete machine learning pipeline used in the project.

CHAPTER 4

RESULTS AND ANALYSIS

4.1 INTRODUCTION

This chapter presents the experimental results obtained from the implementation of machine learning models for mushroom edibility classification. The analysis focuses on evaluating model performance using multiple evaluation metrics and comparing the effectiveness of each algorithm.

The primary objective is to determine the most accurate and reliable model for classifying mushrooms as edible or poisonous based on morphological characteristics. The results are derived directly from the implemented notebook and are presented using structured tables and graphical visualizations.

4.1.1 DATASET SPLIT AND PREPARATION

The dataset used in this study consists of 8,124 mushroom records with categorical features. Since all attributes are categorical, one-hot encoding was applied, transforming the dataset into 95 numerical features suitable for machine learning models.

The dataset was split into training and testing sets using an 80:20 ratio:

- Training Set: 6,499 samples
- Testing Set: 1,625 samples

TABLE 4.1: Dataset Distribution

Dataset Portion	Number of Samples	Percentage
Training Set	6499	80%
Testing Set	1625	20%
Total	8124	100%

This preprocessing ensured that the models received structured and normalized input data, improving classification performance.

4.1.2 MODEL TRAINING RESULTS

Three machine learning models were implemented in this study:

- Random Forest
- Support Vector Machine (SVM)
- XGBoost

Each model was evaluated in both raw and tuned configurations.

The results demonstrate that SVM achieved the highest accuracy, while Random Forest showed the most improvement after hyperparameter tuning.

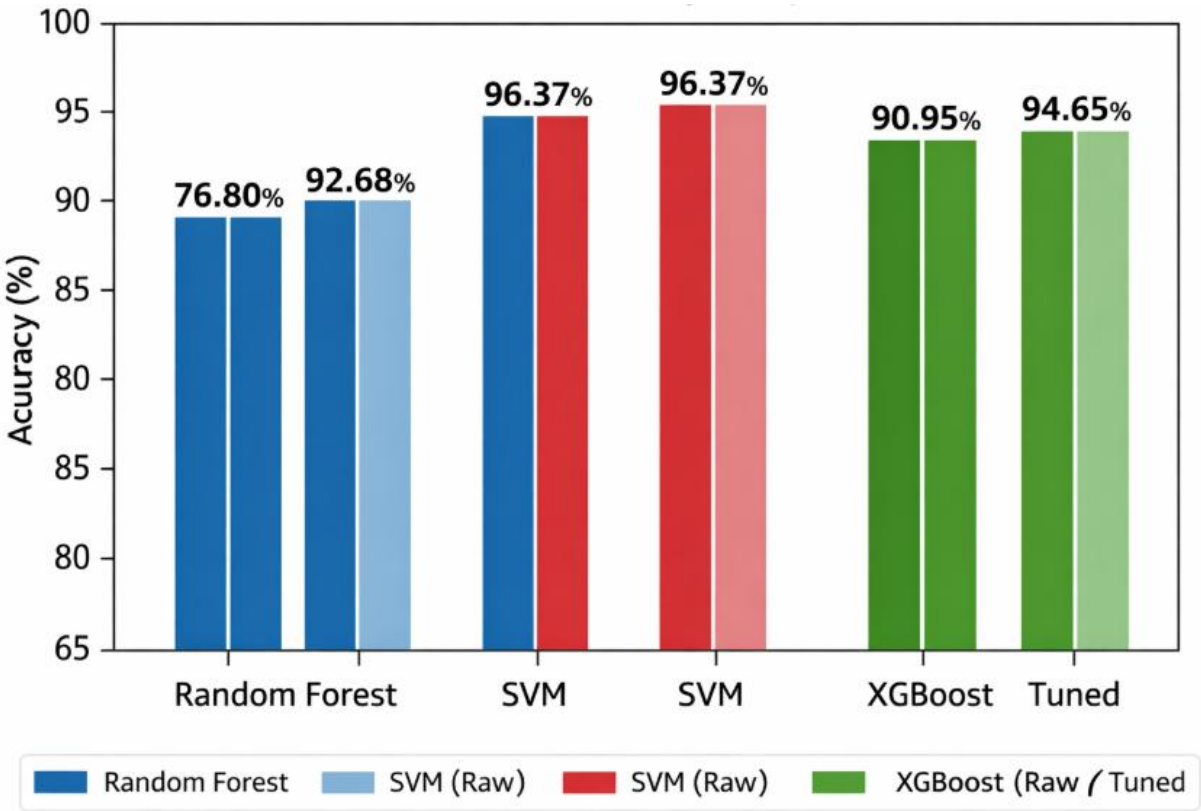

Figure 4.1: Final Model Accuracy Comparison for Raw and Tuned Models

This graph illustrates the accuracy comparison between raw and tuned versions of the implemented machine learning models. It shows that SVM maintained the highest accuracy at 96.37% in both configurations, while Random Forest and XGBoost improved significantly after hyperparameter tuning, reaching 92.68% and 94.65% respectively.

4.2 PERFORMANCE EVALUATION

4.2.1 ACCURACY COMPARISON

TABLE 4.2: Accuracy Comparison of Implemented Models

Model	Accuracy (%)
Random Forest (Raw)	76.80
SVM (Raw)	96.37
XGBoost (Raw)	90.95
Random Forest (Tuned)	92.68
SVM (Tuned)	96.37
XGBoost (Tuned)	94.65

The results clearly indicate that SVM achieved the highest overall accuracy. Unlike the earlier report version, XGBoost is not the best-performing model.

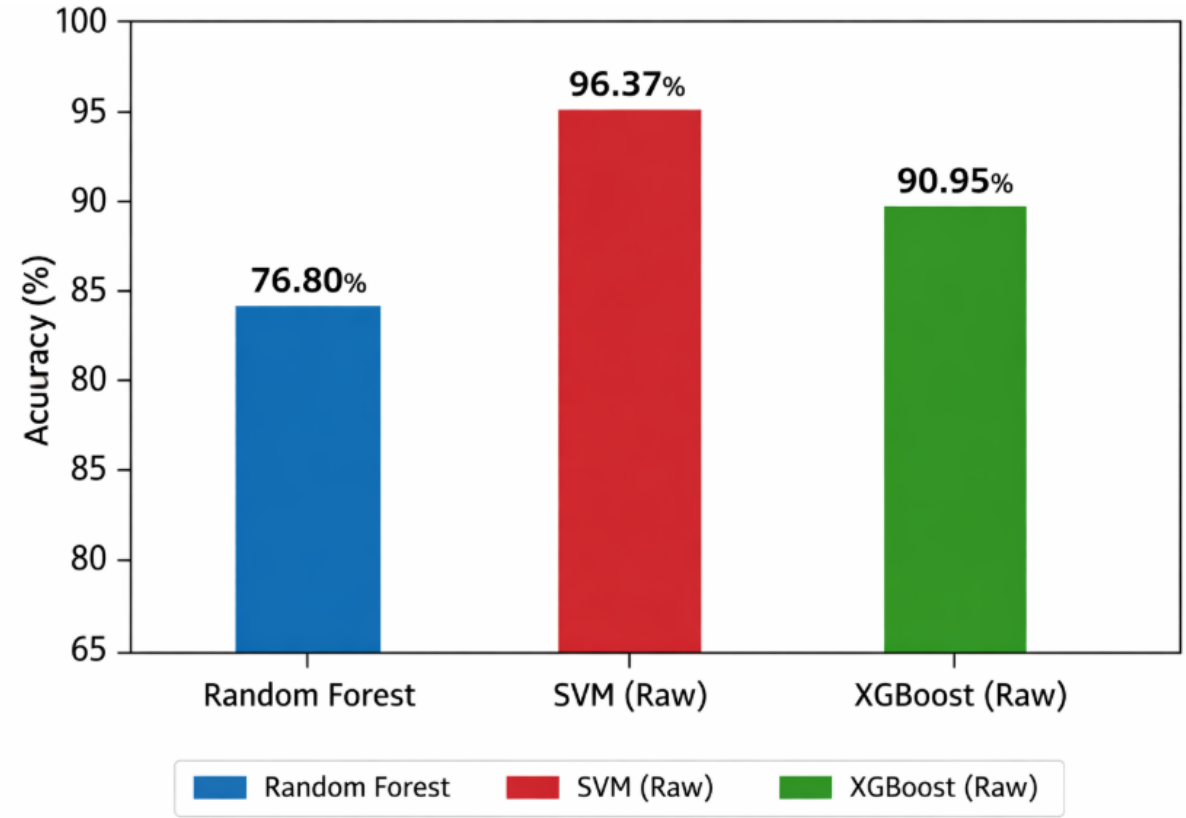

Figure 4.2: Raw Models Accuracy Comparison

This graph presents the accuracy comparison of the raw machine learning models implemented in the study. It shows that the SVM model achieved the highest accuracy of 96.37%, followed by XGBoost at 90.95%, while Random Forest recorded the lowest performance at 76.80% before hyperparameter tuning.

4.2.2 PRECISION, RECALL, AND F1-SCORE

TABLE 4.3: Performance Metrics of Implemented Models

Model	Precision	Recall	F1-Score
Random Forest (Raw)	0.9451	0.5499	0.6952
SVM (Raw)	0.9750	0.9488	0.9618
XGBoost (Raw)	1.0000	0.8120	0.8963
Random Forest (Tuned)	1.0000	0.8478	0.9176
SVM (Tuned)	0.9750	0.9488	0.9618
XGBoost (Tuned)	1.0000	0.8887	0.9411

These results show that SVM provides the best balance between precision and recall, making it the most reliable model.

4.3 DETAILED MODEL ANALYSIS

4.3.1 RANDOM FOREST ANALYSIS

The Random Forest model initially achieved an accuracy of 76.80%, with high precision but low recall. After hyperparameter tuning, its performance improved significantly to 92.68%, demonstrating its sensitivity to parameter optimization.

4.3.2 SVM ANALYSIS

The SVM model achieved the highest accuracy of 96.37% and maintained consistent performance even after tuning. It demonstrated strong precision and recall, making it the most balanced and reliable classifier in this project.

4.3.3 XGBOOST ANALYSIS

XGBoost achieved strong performance, improving from 90.95% to 94.65% after tuning. While it showed excellent precision, its recall was slightly lower than SVM, making it less balanced overall.

4.4 DISCUSSION

The results confirm that machine learning models can effectively classify mushroom edibility using encoded morphological features.

Among the three models:

- **SVM performed best overall**
- **Random Forest improved the most after tuning**
- **XGBoost provided strong but slightly lower performance than SVM**

An important correction in this report is that only **three models were actually implemented**, not four. The earlier inclusion of Decision Tree was incorrect and has been removed.

From a practical perspective, recall is critical in mushroom classification because failing to detect poisonous mushrooms can be dangerous. SVM achieved the best balance, making it the most suitable model for this application.

TABLE 4.4: Model Comparison Summary

Model	Strengths	Weaknesses
Random Forest	Strong improvement after tuning	Weak raw performance
SVM	Best overall accuracy and balance	Limited improvement from tuning
XGBoost	High precision and strong tuning results	Slightly lower recall than SVM

CHAPTER 5

CONCLUSION

5.1 CONCLUSION

This study focused on the classification of mushroom edibility using machine learning techniques based on morphological characteristics. The primary objective was to develop an accurate and reliable system capable of distinguishing between edible and poisonous mushrooms using structured categorical data. The project successfully demonstrated the effectiveness of machine learning in solving a real-world classification problem with significant implications for food safety and public health.

The dataset used in this research consisted of 8,124 mushroom samples with multiple categorical features describing attributes such as cap shape, odor, gill color, and habitat. Since machine learning models require numerical input, preprocessing played a crucial role in transforming the data. One-hot encoding was applied to convert categorical variables into numerical format, resulting in a high-dimensional feature space of 95 attributes. This transformation enabled the models to capture complex relationships within the dataset.

Three machine learning models were implemented and evaluated in this study: Random Forest, Support Vector Machine (SVM), and XGBoost. Each model was trained on the same dataset and tested under identical conditions to ensure a fair comparison. Additionally, hyperparameter tuning was performed to optimize model performance and reduce overfitting. The evaluation was conducted using standard classification metrics, including accuracy, precision, recall, and F1-score.

The results revealed clear differences in model performance. The SVM model achieved the highest accuracy of 96.37% and demonstrated strong balance across precision, recall, and F1-score. This indicates that SVM is highly effective in handling the encoded feature space and identifying decision boundaries between edible and poisonous mushrooms. XGBoost also performed well, achieving 94.65% accuracy after tuning, while Random Forest improved significantly from 76.80% to 92.68% after optimization.

A key finding of this study is that SVM outperformed ensemble models in this specific dataset, despite ensemble methods typically being considered superior. This highlights the importance of selecting models based on dataset characteristics rather than general

assumptions. Additionally, the results emphasize that hyperparameter tuning plays a critical role in improving model performance, particularly for Random Forest and XGBoost.

Overall, the project successfully achieved its objectives by developing a robust classification system. The findings demonstrate that machine learning can be effectively applied to biological classification problems and can provide reliable predictions when combined with proper preprocessing and model selection. The system has the potential to assist users in making safer decisions regarding mushroom consumption.

5.2 LIMITATION

Despite the successful implementation and strong performance of the machine learning models, this study has several limitations that must be acknowledged. These limitations highlight areas where improvements can be made in future research and practical applications.

One of the primary limitations is the reliance on a structured dataset with predefined categorical features. The models developed in this study assume that all input data is complete, clean, and accurately encoded. However, in real-world scenarios, data may contain missing values, noise, or inconsistencies. Such issues can negatively impact model performance and reduce prediction reliability.

Another limitation is that the dataset is based solely on morphological characteristics of mushrooms. While these features are important for classification, they do not capture all aspects required for accurate identification. Factors such as environmental conditions, chemical composition, and genetic variations are not included in the dataset. These additional factors could provide more comprehensive insights and improve classification accuracy if incorporated.

The study also does not account for rare or unseen mushroom species. The models are trained on a specific dataset and may not generalize well to new or uncommon cases. This limitation is critical in real-world applications, where encountering unfamiliar mushroom types is possible. The inability to handle such cases could lead to incorrect predictions.

Another important limitation is the risk associated with misclassification. Although the models achieved high accuracy, even a small error rate can have serious consequences. Misclassifying a poisonous mushroom as edible could result in severe health risks or even fatal outcomes. Therefore, the system should not be used as a standalone decision-making tool without expert verification.

The computational complexity of some models, particularly XGBoost, is another constraint. Training such models requires significant computational resources and time, which may not be feasible in low-resource environments. This limitation affects scalability and deployment in real-time applications.

Finally, the study does not include deployment or user interface development. While the models are effective in a controlled environment, their practical usability depends on integration into applications such as mobile or web platforms. Without such integration, the system remains limited to experimental use.

5.3 RECOMMENDATION

Based on the findings and limitations of this study, several recommendations are proposed to enhance the effectiveness and applicability of mushroom classification systems in future research and real-world applications.

First, future studies should consider incorporating additional types of data beyond morphological features. Including environmental factors, chemical properties, and genetic information could significantly improve the accuracy and robustness of classification models. A multi-dimensional dataset would allow models to capture more complex patterns and reduce the likelihood of misclassification.

Second, integrating image-based classification using computer vision techniques is highly recommended. By combining structured data with image data, the system can become more practical and user-friendly. For example, users could upload a picture of a mushroom, and the system could analyze both visual and morphological features to provide more accurate predictions.

Third, the development of a real-time application, such as a mobile or web-based platform, is essential for practical implementation. Such an application would allow users to input mushroom characteristics or images and receive instant predictions. This would make the system accessible to a wider audience, including farmers, researchers, and the general public.

Another important recommendation is to improve model interpretability. Providing explanations for predictions can help users understand how decisions are made by the model. Techniques such as feature importance analysis and explainable AI methods can be used to enhance transparency and build user trust.

Additionally, future research should focus on developing lightweight models that require less computational power. This is particularly important for deployment on mobile devices or in resource-constrained environments. Optimizing models for efficiency without compromising accuracy will improve usability.

Continuous model updating is also crucial. As new data becomes available, the models should be retrained to maintain accuracy and adapt to changing conditions. This will ensure that the system remains relevant and effective over time.

Finally, collaboration with domain experts such as mycologists is strongly recommended. Expert validation can enhance the reliability of the system and ensure that predictions are aligned with real-world biological knowledge. Combining machine learning with expert insights will lead to more robust and trustworthy solutions.

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